



Cataract Surgery Postoperative Care

M. Hermina Strungaru, MD, PhD, FRCSC
Meghan Smith, MD

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Introduction

Without proper tools (such as a visual acuity chart, slit lamp or indirect ophthalmoscope), it is difficult for a primary care practitioner to diagnose and manage the complications of a postoperative cataract patient. The goal of this chapter is to review signs and symptoms that would require an urgent referral to an ophthalmologist. These include postoperative infection, glaucoma, inflammation or any suspected issues with healing. [Figure 1](#) can be utilized to help assess if urgent referral is required.

The medications listed within the chapter are those that are typically given *prophylactically* to prevent infection (see [Table 2](#)), a rise in intraocular pressure (IOP) (see [Table 3](#)) and cystoid macular edema (CME) (see [Table 4](#)), as well as to improve healing (see [Table 5](#)).

Goals of Therapy

The goals of *perioperative prophylactic pharmacologic treatment* are:

- Control inflammation
- Prevent infection
- Maintain eye comfort
- Promote early visual rehabilitation

The goals of *postoperative assessment* are:

- Detect intraocular infection in its early stages
- Detect postoperative uveitis or IOP elevation

- Detect other abnormalities in the postoperative course, such as a retinal detachment, iris prolapse, wound leak, flat anterior chamber, excessive corneal edema or intraocular hemorrhage

Investigations

[Figure 1](#) illustrates an assessment process for the postoperative cataract patient. [Table 1](#) summarizes the red flags requiring urgent referral.

- Pain:
 - the postoperative eye should be comfortable; at worst, the patient may have a mild foreign-body sensation
 - more intense pain may suggest increased IOP, increased inflammation and/or infection; postoperative inflammation is more common within the first 48 hours of surgery, while postoperative infection is most common between 3 and 7 days following surgery
- History of recent trauma:
 - any trauma to the eye in the early postoperative phase requires thorough reassessment with a slit lamp
- Change in vision (worsening, darkening, loss of detail or peripheral visual loss):
 - any significant change could indicate infection, hemorrhage, retinal detachment or other acute intraocular pathology requiring immediate attention
- Visual phenomena (flashing lights, dark shadows or floaters):
 - worsening of floaters requires thorough reassessment to rule out infection, retinal tear/detachment or uveitis
 - in the first few days or weeks after implantation of an intraocular lens, many patients notice a glare or shadow in the peripheral vision (dysphotopsia); if within this time frame, it is usually not a cause for concern, provided that confrontation visual field testing (finger counting using peripheral vision) is normal
 - using a pinhole to test visual acuity will eliminate the effect of refractive error if the patient has apparent reduced acuity
- Itching of the eye (as predominant symptom):
 - suggests allergy to medications or dry eye
 - patient should be re-examined to determine the cause of itching and whether alternate eye drops should be prescribed
- Examination of eye:
 - swelling of lids and/or conjunctiva suggest drug allergy or infection
 - pupil(s) should react normally to light unless a mydriatic agent has been used; photophobia (glare sensitivity and pain from light exposure) can indicate anterior uveitis or corneal haze (from infection, inflammation or increased IOP)

- the cornea should be clear of any clouding or infiltrates (to rule out corneal or anterior segment infection, inflammation, or IOP elevation); it should reflect a clear, well-demarcated image when the direct ophthalmoscope or pupil light is shone on it
- ophthalmoscope should be used to see if the disc is clearly viewed and a red reflex is present (to rule out vitreous clouding, inflammation or infection)
- Review of ocular medications:
 - reinforce proper use
 - clarify any confusion
 - discuss with family member
- Verify follow-up visits with surgeon

Table 1: Red Flags Requiring Urgent Referral

On history:	On examination:	
Trauma to eye postsurgery	Swollen or red eyelids or conjunctiva	Hazy cornea
Blurry or worsening vision	Hypopyon (white layer) in anterior chamber	Poor or absent red reflex with direct ophthalmoscopy
Persistent or progressively painful eye (may also have headache, nausea, vomiting or feel unwell) or light sensitivity	Flat (poorly formed) anterior chamber	Seidel-positive wounds (leakage of fluid from anterior chamber when fluorescein is instilled in eye) or iris peaking toward wound or prolapsing from wound
Persistent or progressive shadows, flashes, or floaters	Elevated intraocular pressure	

Therapeutic Choices

Although cataract surgery is a very commonly performed procedure, practice patterns related to patients' preoperative and postoperative care vary depending upon the surgeons' preferences and habits.

Pharmacologic Choices

Prophylactic ophthalmic medications may include:

- **Antibacterials** (see [Table 2](#))

The measures used to reduce the risk of postoperative endophthalmitis are preoperative use of iodized povidone, preoperative and postoperative antibiotics, and the use of intraoperative intracameral antibiotics.^{[1][2]}

A broad-spectrum antibacterial is often used perioperatively to reduce the risk of endophthalmitis. The endophthalmitis rates reported in the literature have decreased from 0.32% in the 1970s to 0.29% or less in the early 2000s, depending upon the country.^{[3][4][5]} Declining rates of infection have been attributed to better technology, perioperative use of fourth-generation fluoroquinolones and the introduction of anterior chamber injection of antibiotics at the end of the surgery.^{[6][7][8]}

Despite inconsistent evidence that perioperative antibiotics reduce the rate of endophthalmitis, antimicrobials before, during and after cataract surgery have become standard of care. However, there are controversies regarding the choice of antimicrobial, its route of administration, duration and timing.^{[9][10]} Perioperative topical antibiotics such as fourth-generation fluoroquinolones (**besifloxacin**, **gatifloxacin**, **moxifloxacin**) are commonly used.^{[11][12]} Alternatively, fixed combination preparations containing an antibacterial and a corticosteroid, e.g., **tobramycin/dexamethasone**, may be preferred by some surgeons for their ease of use, which can potentially improve adherence to therapy. Local bacterial resistance patterns may also influence a surgeon's choice of perioperative antibacterial agent.^{[13][14][15][16]}

Alternatively, an antibiotic can be administered by intracameral injection, whereby the antibiotic is injected directly into the anterior chamber of the eye at the end of the surgery. Evidence for the effectiveness of this practice is conflicting. Some studies have shown no benefit with this method,^{[17][18]} while others, including one study by the European Society of Cataract and Refractive Surgeons, have found it to be more effective than intraoperative topical antibacterial therapy.^{[6][19][20][21][22][23][24]} Furthermore, there are concerns regarding intracameral use of noncommercially prepared antibiotic preparations, which may impact a surgeon's decision.^[25] These include issues such as development of resistant strains or toxic anterior segment syndrome.^{[13][14][15][16][26]} Most recently, the use of intracameral **vancomycin** was associated with a rare but devastating complication termed hemorrhagic occlusive retinal vasculitis (HORV).^[27] When selecting an intracameral agent, the efficacy and cost-effectiveness of an agent may guide a surgeon's decision.^[23]

- **Glaucoma medications** (see [Table 3](#))

These medications are used temporarily to lower the IOP after eye surgery if it is elevated. Patients with pre-existing glaucoma should be encouraged to continue using their preoperative glaucoma drops. The medication regimen may need to be modified during the postoperative period to protect against an early postoperative rise in IOP. If an IOP spike occurs in the postoperative period, it generally lasts only a few days and is amenable to topical medical management. The surgeon will typically select a noninflammatory antiglaucoma agent, such as an alpha-adrenergic agent, beta-adrenergic antagonist or carbonic anhydrase inhibitor, combined with close monitoring to ensure pressure-lowering effect. Occasionally a surgical procedure called an anterior chamber paracentesis may be required to reduce the pressure.

- **Anti-inflammatory agents** (see [Table 4](#))

These medications are used for a few weeks postoperatively to reduce inflammation and the risk of developing CME. A Cochrane review of low-quality evidence demonstrated that **ophthalmic NSAIDs** alone or in combination with **ophthalmic corticosteroids** may be more effective in controlling inflammation and preventing CME when compared with ophthalmic corticosteroids alone.^[28] Additional studies have validated the efficacy of topical NSAIDs in the prevention of postoperative inflammation and CME.^{[29][30][31]} Anti-inflammatory agents may contribute to a faster visual recovery and reduced postoperative pain.^[32] There is variation in practice regarding the selection of anti-inflammatory agent, so it is best to consult the surgeon if there is any confusion regarding which drop the patient should use.^{[33][34][35][36][37][38][39]}

- **Ophthalmic dilators** and **cycloplegics** (see [Table 5](#))

These medications are used to keep the iris away from the implant during the early healing period and to improve comfort by decreasing ciliary muscle spasm.

- **Oral analgesics** may be required (e.g., **acetaminophen** 500 mg every 4–6 hours).

Other systemic medications should be continued, including antiplatelets and anticoagulants, unless otherwise specified by the surgeon.

Therapeutic Tips

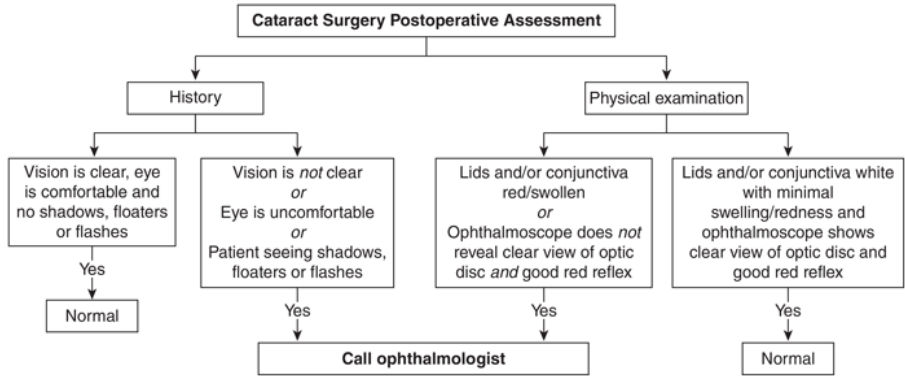
- Any changes to postoperative ophthalmic medications should be discussed with the treating ophthalmologist.
- Initiate topical antibacterial drops immediately following surgery rather than waiting until the first postoperative day.^{[40][41][42]}
- Advise patients to separate the administration of different eye drops by a period of at least 5 minutes. If there is a contraindication to systemic absorption of the medication, counsel the patient to close the eye and, while trying to avoid touching the operated eye, put pressure on the inner canthus for 30–60 seconds after instilling drops. This is done to reduce the transfer of ophthalmic medication to the nasal and/or oral mucosa where it may be absorbed systemically. In all other patients, simply closing the eye for 30–60 seconds may be sufficient to maximize ophthalmic absorption.
- Any worsening of vision, floaters, eye pain or redness, especially in the first postoperative week, should be considered endophthalmitis until proven otherwise and requires urgent assessment by an ophthalmologist.
- Treatment with many medications in this setting is for a limited course; therefore, any unused ophthalmic medication should be disposed of properly.
- Patients having clear corneal cataract surgery should take all their usual medications (including anticoagulants and antiplatelet agents) on the day of surgery except for some diabetic medications. Due to the risk of hypoglycemia, insulin and insulin secretagogue (sulfonylureas, meglitinides) doses may be modified or the medication may be held the day before or on the day of surgery. Metformin may be held the day before and on the day of surgery due to the risk of lactic acidosis. If medication doses and/or

regimens are modified, more frequent glucose monitoring may be required.^[43] This should be discussed with the physician prior to surgery.

- There may be room for improvement in prescribing patterns, with potential substantial cost savings, given the lack of superior efficacy of costly versus less-costly eye drops in the perioperative period.^[44]

Algorithms

Figure 1: Evaluation of the Postoperative Cataract Patient by the Primary Care Practitioner



Drug Tables

Table 2: Antibacterials for Cataract Surgery Postoperative Care

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
Drug Class: Aminoglycosides			
tobramycin 0.3% Tobrex , generics	1 drop in affected eye BID to QID	Extended use may cause conjunctivitis or epitheliopathy reactions in the eye.	High concentrations on and in the eye lead to broader bacterial coverage than traditional in vitro testing suggests.
< \$10	× 7–10 days		

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
			Preservative is benzalkonium chloride.
Drug Class: Aminoglycoside Combinations			
neomycin/dexamethasone/polymyxin B sulfate Maxitrol \$10–20	1–2 drops in affected eye 4–6 times daily × 7–10 days	Extended use may cause conjunctivitis or epitheliopathy reactions in the eye. Dexamethasone: elevated intraocular pressure; anti-inflammatory effects can mask signs of infection.	High concentrations on and in the eye lead to broader bacterial coverage than traditional in vitro testing suggests. Dexamethasone: commonly used after all types of eye surgery. Preservative is benzalkonium chloride.
Drug Class: Fluoroquinolones			
besifloxacin 0.6% Besivance \$10–20	1 drop in affected eye TID × 7–10 days	Blurred vision, local irritation or discomfort.	Fourth-generation fluoroquinolone; improved broad-spectrum activity, especially against gram-positive organisms. Preservative is benzalkonium chloride.

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
ciprofloxacin 0.3% Ciloxan , generics < \$10	1 drop in affected eye BID to QID × 7–10 days	Formation of white crystalline precipitate on a corneal defect after administration.	Excellent broad-spectrum profile and minimal ocular surface toxicity; very commonly used; may be used orally if lens capsule ruptured during cataract surgery. Preservative is benzalkonium chloride.
ciprofloxacin , oral Cipro , generics \$10–20	500 mg Q12H PO × 7–10 days	Abdominal pain, headache, dizziness, photosensitivity, hepatitis, cartilage toxicity, pseudomembranous colitis.	Used orally if lens capsule ruptured during cataract surgery.
gatifloxacin 0.3% Zymar , generics \$10–20	1 drop in affected eye BID to QID × 7–10 days	Blurred vision, local irritation or discomfort. Well tolerated and minimal eye toxicity with topical administration. Allergic reactions uncommon.	Fourth-generation fluoroquinolone; improved broad-spectrum activity, especially against gram-positive organisms. Preservative is benzalkonium chloride.

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
moxifloxacin 0.5% Vigamox, generics \$10–20	1 drop in affected eye QID × 7–10 days	Blurred vision, local irritation or discomfort. Well tolerated and minimal eye toxicity with topical administration. Allergic reactions uncommon.	Fourth-generation fluoroquinolone; improved broad-spectrum activity, especially against gram-positive organisms. Preservative-free.
ofloxacin 0.3% Ocuflox \$10–20	1 drop in affected eye QID × 7–10 days	Blurred vision, local irritation or discomfort. Well tolerated and minimal eye toxicity with topical administration. Allergic reactions uncommon.	Excellent broad-spectrum profile and minimal ocular surface toxicity; very commonly used; may be used orally if lens capsule ruptured during cataract surgery. Preservative is benzalkonium chloride.

^[a] Cost of smallest available pack size or 10-day supply for oral treatment; includes drug cost only.
👤 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Table 3: Glaucoma Medications for Cataract Surgery Postoperative Care

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
Drug Class: Alpha ₂ -adrenergic Agonists, ophthalmic			
apraclonidine 1.0% lopidine \$20–30	1 drop in affected eye BID Used as long as intraocular pressure is elevated, or permanently in	Common ocular reactions include hyperemia and a burning sensation. Other reactions include dry mouth, tachycardia, hypotension, headache, tremor.	Potent in controlling postoperative pressure spikes; avoid in patients with severe cardiovascular disease in light of alpha-adrenergic effects; use with caution in patients taking tricyclic antidepressants.

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
Drug Class: Beta-adrenergic Antagonists, ophthalmic			
brimonidine 0.2% Alphagan, generics < \$10	patients with pre-existing glaucoma		Contraindicated with MAO inhibitors. Preservative is benzalkonium chloride.
	1 drop in affected eye BID Used as long as intraocular pressure is elevated, or permanently in patients with pre-existing glaucoma	Common ocular reactions include hyperemia and a burning sensation. Other reactions include dry mouth, tachycardia, hypotension, headache, tremor.	Potent in controlling postoperative pressure spikes; avoid in patients with severe cardiovascular disease in light of alpha-adrenergic effects; use with caution in patients taking tricyclic antidepressants. Contraindicated with MAO inhibitors. Preservative is benzalkonium chloride.
brimonidine 0.15% Alphagan P, generics < \$10	1 drop in affected eye TID Used as long as intraocular pressure is elevated, or permanently in patients with pre-existing glaucoma	Common ocular reactions include hyperemia and a burning sensation. Other reactions include dry mouth, tachycardia, hypotension, headache, tremor.	Potent in controlling postoperative pressure spikes; avoid in patients with severe cardiovascular disease in light of alpha-adrenergic effects; use with caution in patients taking tricyclic antidepressants. Contraindicated with MAO inhibitors. Preservative is purite; may have slightly lower incidence of ocular allergy than brimonidine 0.2%.
Drug Class: Beta-adrenergic Antagonists, ophthalmic			
betaxolol 0.25% Betoptic S	1 drop in affected eye BID	Symptoms of systemic beta blockade can be precipitated or aggravated	Efficacious and cost effective with minimal ocular toxicity; avoid in

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
\$10–20	Used as long as intraocular pressure is elevated, or permanently in patients with pre-existing glaucoma	with beta-blocking eye drops; potentially serious systemic effects in patients with asthma, obstructive lung disease or borderline cardiac function. Local adverse effects (minimal): stinging, dry eye, conjunctivitis (rare).	asthmatics or patients with obstructive lung disease or borderline cardiac function. Preservative is benzalkonium chloride.
timolol 0.5% Timoptic, Apo-Timop, other generics < \$10	1 drop in affected eye BID Used as long as intraocular pressure is elevated, or permanently in patients with pre-existing glaucoma	Symptoms of systemic beta blockade can be precipitated or aggravated with beta-blocking eye drops; potentially serious systemic effects in patients with asthma, obstructive lung disease or borderline cardiac function. Local adverse effects (minimal): stinging, dry eye, conjunctivitis (rare).	Efficacious and cost effective with minimal ocular toxicity; avoid in asthmatics or patients with obstructive lung disease or borderline cardiac function. Preservative is benzalkonium chloride.
timolol 0.5% gel-forming solution Timoptic XE, generics \$10–20	1 drop in affected eye QAM Used as long as intraocular pressure is elevated, or permanently in patients with pre-existing glaucoma	Symptoms of systemic beta blockade can be precipitated or aggravated with beta-blocking eye drops; potentially serious systemic effects in patients with asthma, obstructive lung disease or borderline cardiac function. Local adverse effects (minimal): stinging, dry eye, conjunctivitis (rare).	Efficacious and cost effective with minimal ocular toxicity; avoid in asthmatics or patients with obstructive lung disease or borderline cardiac function. Once-daily formulation improves adherence. Shake container once prior to each use. Preservative is benzododecinium bromide.
Drug Class: Carbonic Anhydrase Inhibitors, ophthalmic			

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
brinzolamide 1% Azopt \$10–20	1 drop in affected eye BID Use may be long term	Stinging; bitter, sour or unusual taste. Conjunctivitis and blepharitis are not uncommon side effects with topical carbonic anhydrase inhibitors.	Preservative is benzalkonium chloride.
dorzolamide 2% Trusopt, generics \$10–20	1 drop in affected eye BID Use may be long term	Stinging; bitter, sour or unusual taste. Conjunctivitis and blepharitis are not uncommon side effects with topical carbonic anhydrase inhibitors.	Preservative in Trusopt is benzalkonium chloride; however, a preservative-free formulation is available.
Drug Class: Prostaglandin Analogues, ophthalmic			
bimatoprost 0.01% or 0.03% Lumigan RC, Vistitan \$40–50	1 drop in affected eye QHS Used as long as intraocular pressure is elevated, or permanently in patients with pre-existing glaucoma	Brown pigmentation of the iris; darkening of the periocular skin; thickening and elongation of eyelashes. Headache, flulike symptoms. Hyperemia not uncommon in first month of use.	Very potent pressure-lowering agents with once-daily application. Preservative is benzalkonium chloride.
latanoprost 0.005% Xalatan , generics < \$10	1 drop in affected eye QHS Used as long as intraocular pressure is elevated, or permanently in patients with pre-existing glaucoma	Brown pigmentation of the iris; darkening of the periocular skin; thickening and elongation of eyelashes. Headache, flulike symptoms. Hyperemia not uncommon in first month of use.	Very potent pressure-lowering agents with once-daily application. Preservative is benzalkonium chloride.
travoprost 0.004% Izba , Travatan Z , generics	1 drop in affected eye QHS	Brown pigmentation of the iris; darkening of the periocular skin; thickening	Very potent pressure-lowering agents with once-daily application.

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
\$10–20	Used as long as intraocular pressure is elevated, or permanently in patients with pre-existing glaucoma	and elongation of eyelashes. Headache, flulike symptoms. Hyperemia not uncommon in first month of use.	Izba: POLYQUAD preservative system (polyquaternium-1). Other products: sofZia preservative system (boric acid, propylene glycol, sorbitol, zinc chloride).
Drug Class: Carbonic Anhydrase Inhibitors, oral			
acetazolamide  , oral generics	250 mg up to QID PO Used for very high IOP on a short-term basis	Paresthesias of the extremities, metabolic acidosis, hypokalemia, kidney stones, gastrointestinal upset, lethargy and depression, aplastic anemia, Stevens-Johnson syndrome.	Carbonic anhydrase inhibitors more commonly used topically as adjunctive agents to reduce intraocular pressure. Oral formulation usually used short term for severe rises in pressure or treatment failure with topical formulations.
\$10–20			
methazolamide, oral generics	50–100 mg BID–TID PO Used for very high IOP on a short-term basis	Paresthesias of the extremities, metabolic acidosis, hypokalemia, kidney stones, gastrointestinal upset, lethargy and depression, aplastic anemia, Stevens-Johnson syndrome. Reduced severity of side effects compared to acetazolamide.	Carbonic anhydrase inhibitors more commonly used topically as adjunctive agents to reduce intraocular pressure. Oral formulation usually used short term for severe rises in pressure or treatment failure with topical formulations.
\$30–40			

^[a] Cost of smallest available pack size or 30-day supply of oral therapy; includes drug cost only.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Abbreviations: IOP = intraocular pressure; MAO = monoamine oxidase

Table 4: Ophthalmic Anti-inflammatories for Cataract Surgery Postoperative Care

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
Drug Class: Corticosteroids, ophthalmic			
dexamethasone 0.1% Maxidex	1–2 drops in affected eye 4–6 times daily	Elevated intraocular pressure; anti-inflammatory effects can mask signs of infection.	Commonly used after all types of eye surgery.
< \$10	× 3–4 wk; longer if evidence of cystoid macular edema		Preservative is benzalkonium chloride.
difluprednate 0.05% Durezol	1 drop in affected eye QID	Elevated intraocular pressure; anti-inflammatory effects can mask signs of infection.	Preservative is sorbic acid.
\$20–30	Start 24 h after surgery, continue × 2 wk then taper		
fluorometholone 0.1% Flarex, FML, generics	1–2 drops in affected eye BID–QID	Elevated intraocular pressure; anti-inflammatory effects can mask signs of infection.	Commonly used after all types of eye surgery.
< \$10	× 3–4 wk; longer if evidence of cystoid macular edema		Less elevation of intraocular pressure compared with dexamethasone or prednisolone, but less potent. Preservative is benzalkonium chloride.

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
	Start 1 day before surgery; continue on day of surgery then × 2 wk postoperatively; longer if evidence of cystoid macular edema	possible ulceration with prolonged and frequent use. Eyelid margin crusting, eye pain, punctate keratitis, blurred vision, dry eye, pruritis, headache.	corticosteroids); reduces risk of developing cystoid macular edema; should not be used in patients with asthma or allergies to ASA or NSAIDs. Shake well before using. Preservative is benzalkonium chloride.

^[a] Cost of smallest available pack size; includes drug cost only.

Abbreviations: ASA = acetylsalicylic acid

Table 5: Ophthalmic Dilators and Cycloplegics for Cataract Surgery Postoperative Care

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
Drug Class: Dilators and Cycloplegics			
cyclopentolate 1% Cyclogyl, Odan-Cyclopentolate, other generics \$10–20	1 drop in affected eye TID × first few weeks after surgery	Systemic anticholinergic side effects, e.g., flushing, tachycardia, urinary retention (uncommon).	Less commonly used with modern cataract surgery in light of smaller incisions and less postoperative inflammation. Preservative is benzalkonium chloride.
phenylephrine 2.5% Mydrin, generics < \$10	1 drop in affected eye TID × first few weeks after surgery	Tachycardia and hypertension (rare).	Less commonly used with modern cataract surgery in light of smaller incisions and less postoperative inflammation. Preservative is benzalkonium chloride.

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
tropicamide 0.5% or 1% Mydracyl, generics \$10–20	1 drop in affected eye TID × first few weeks after surgery	Systemic anticholinergic side effects, e.g., flushing, tachycardia, urinary retention (uncommon).	Less commonly used with modern cataract surgery in light of smaller incisions and less postoperative inflammation. Preservative is benzalkonium chloride.

^[a] Cost of smallest available pack size; includes drug cost only.

Suggested Readings

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